

Analysis group meeting

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- Cooking for pass0v10 is done.
- Only 3 days for the cooking, it was fast.
- Detector timelines are published.

- Working on the high-level physics QA timelines.
 - I need to submit jobs with slurm
 - These jobs are generated by the script `run-monitoring.sh`

- Still working on the Vz pull request, I'm working on displaying the two means.

ALERT pull request

The fix for the generation of the track candidates is done.

ALERT: Improvement made to the algorithm for generating all track candidates. #508

 Merged baltzell merged 6 commits into [JeffersonLab:development](#) from [mathieuouillon:development](#)  4 days ago

 Conversation 3  Commits 6  Checks 33  Files changed 5



mathieuouillon commented 5 days ago

Collaborator ...

- Change the algorithm to generate all track candidates for better readability.
- Add a maximum number of track candidates generated to limit the time and memory taken by this function.
- Change the option to select the track finding to an enum.

Now, this option can be set up from the YAML file with the option Mode:

```
AHDC:
  Mode: "AI_Track_Finding" # Default is CV_Track_Finding
```



- Add a cut on the number of hits to rely on the conventional track finding with more than 300 hits.
- Change the default options for DJL:
 - Use a single thread for the inference.
 - Remove warning about the graph executor optimization



ALERT pull request

Pull request by Eric have been merged also.

New features for ALERT Kalman filter+optimized parameters #445

 Merged baltzell merged 44 commits into [JeffersonLab:development](#) from [efuchey:development](#) 4 days ago

 Conversation 1  Commits 44  Checks 33  Files changed 10

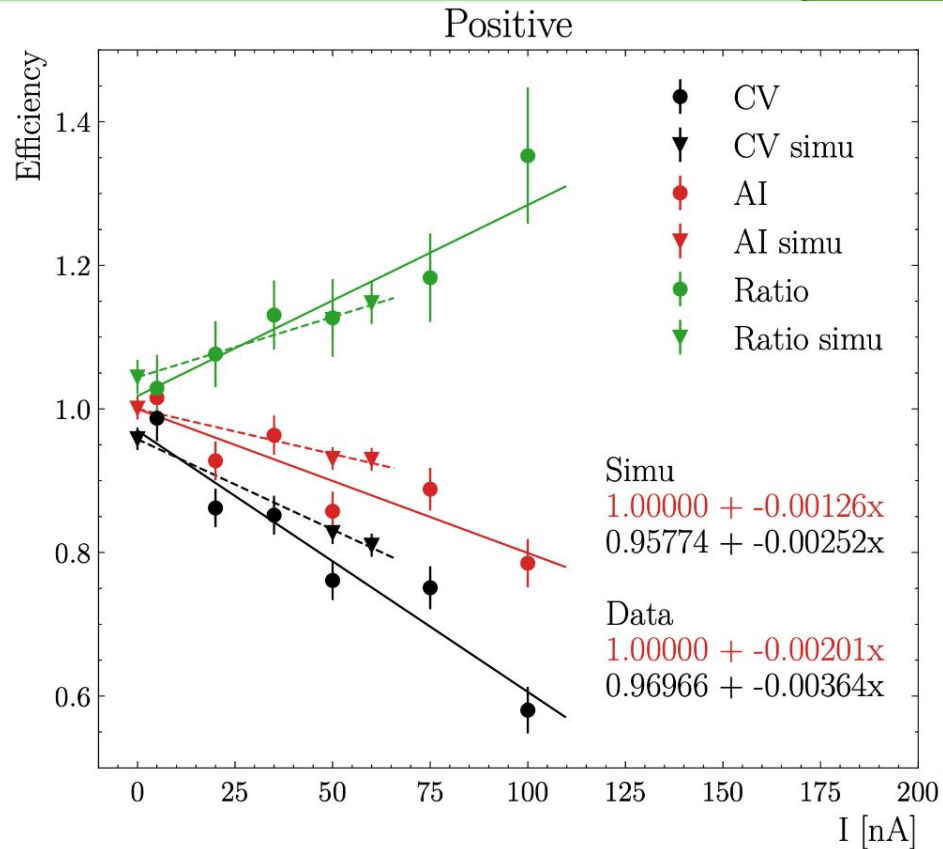
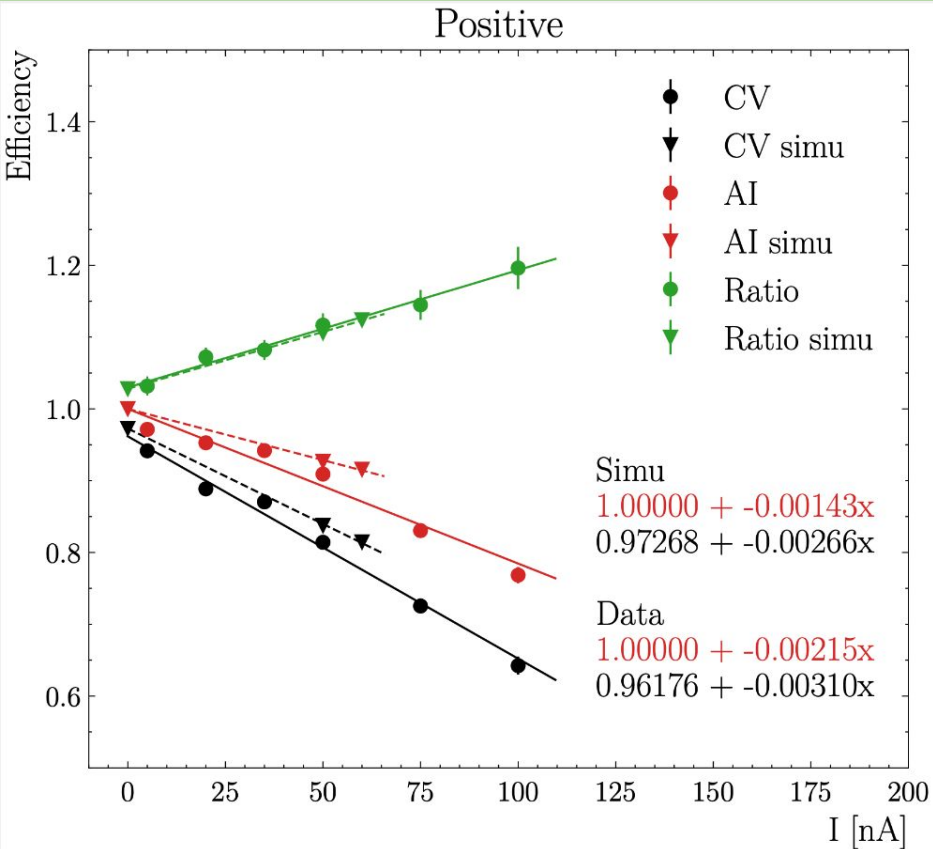


efuchey commented on Jan 27 • edited ▾

Contributor ...

- I have redone the simulation files for the background merging with inbending setting.
- gcard: /work/clas12/ouillon/background_merging/clas12_1D2-RGD_inbending.gcard
- Do you want to check the gcard now on I can send you the gcard by email?

RG-D: Background merging

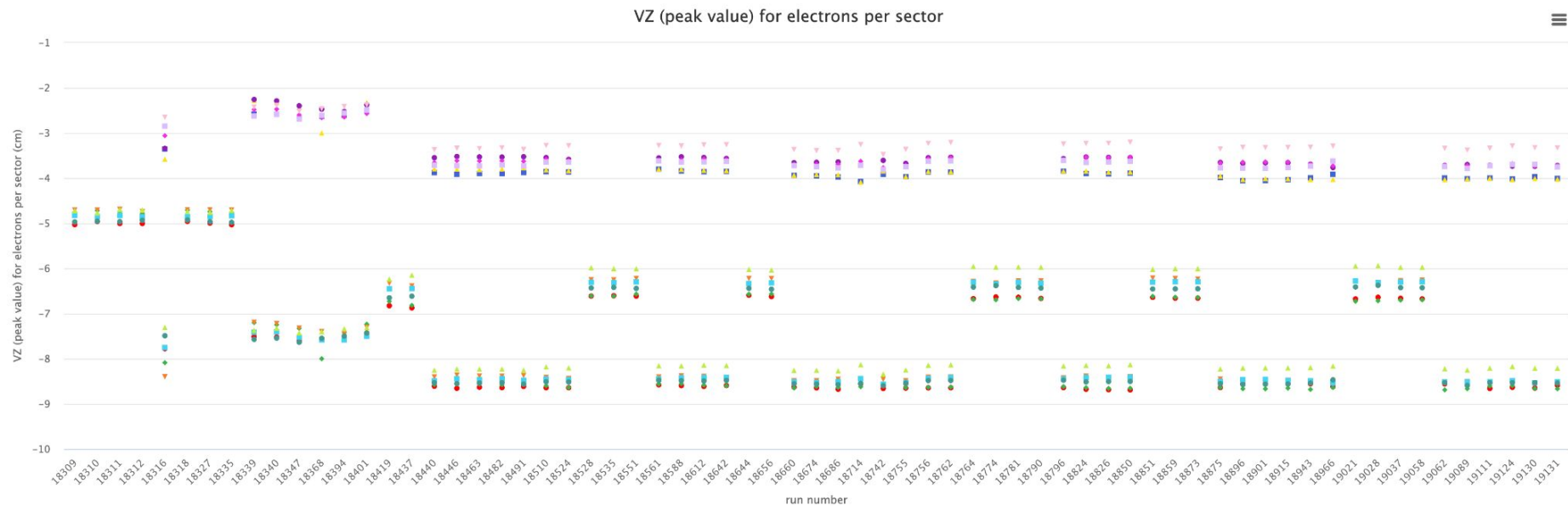


- I show this plot to Raffaella.
- She told me the plots look good.
- We should expect the slope of the simulation to be smaller than the data.
- Raffaella told me that it could be a good idea to do the same thing for a nuclear target. I think carbon is the simplest nuclear target.

Timelines

The two means are plotted.

I need to improve the code (variable name + change negative and positive timelines) and make a pull request.



Cuts (for black distribution):

Electron:

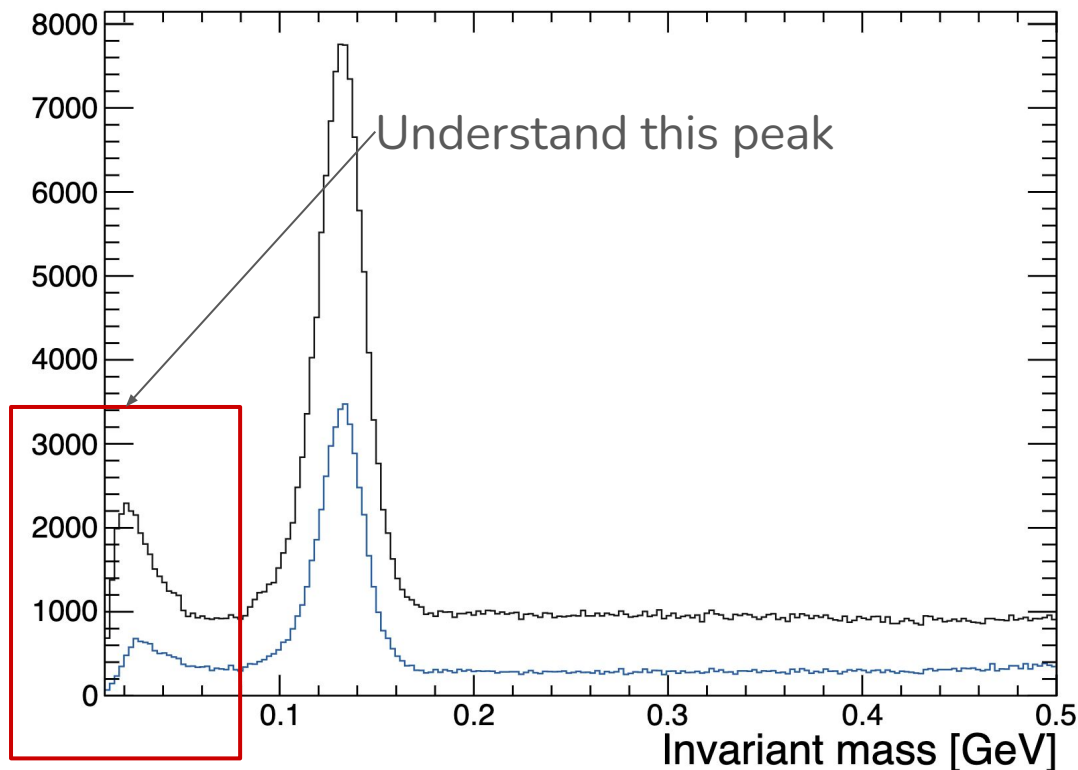
- $\text{pid} = 11 \ \&\& \ \text{status} < 0$
- $-3 < \chi^2_{\text{pid}} < 3$
- $-15 < V_z < 5 \text{ cm}$

Photon:

- $E > 0.6 \text{ GeV}$
- $1.57077 < \Delta t < 1.68663 \text{ ns}$
- $\theta_{e\gamma} > 8 \text{ deg}$

Cuts (for the blue distribution):

- $Q^2 > 1 \text{ GeV}^2$
- $W > 2 \text{ GeV}$
- $0.3 < z_h < 0.7$



Cuts (for black distribution):

Electron:

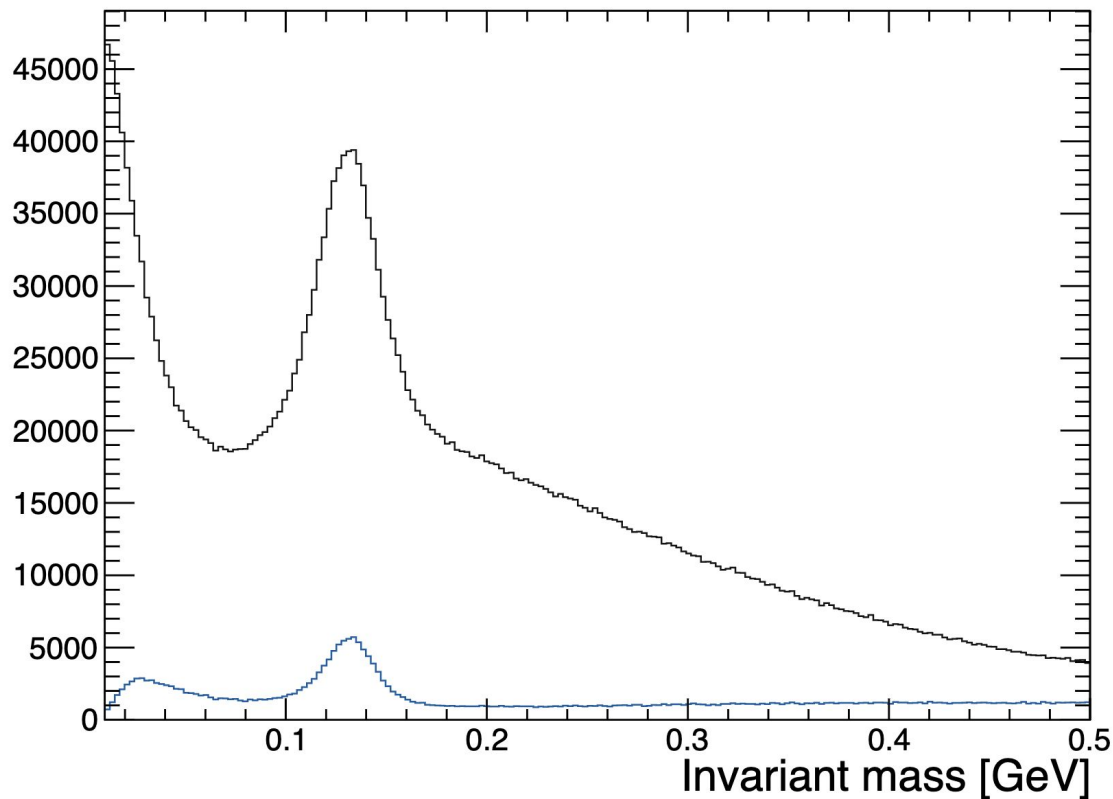
- `pid = 11 && status < 0`
- `-3 < chi2pid < 3`
- `-15 < Vz < 5 cm`

Photon:

- `E > 0.2 GeV`

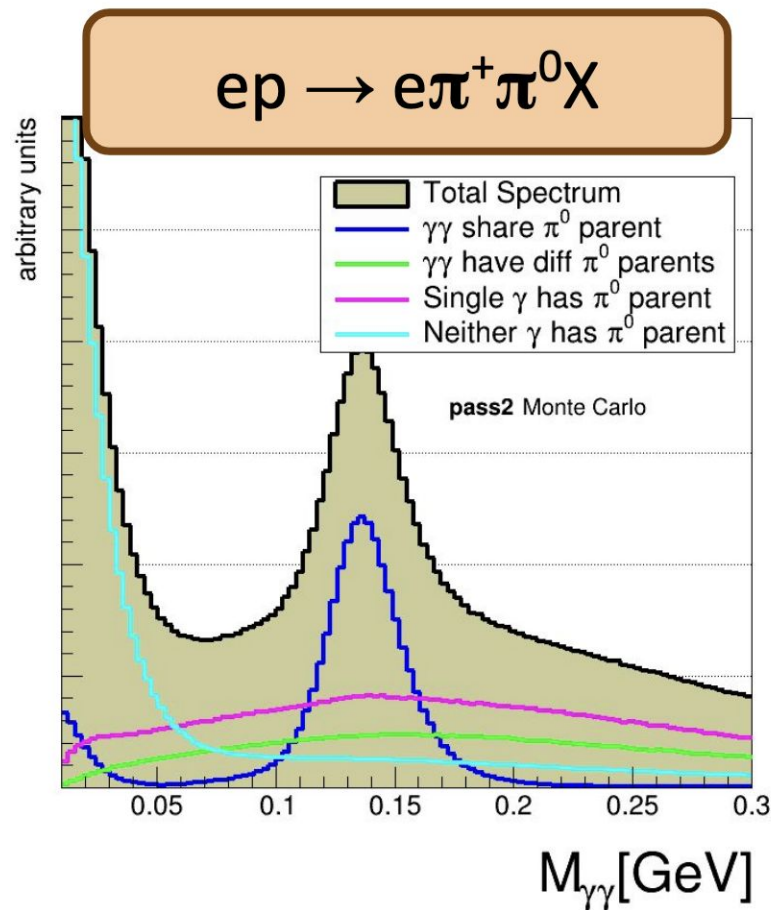
Cuts (for the blue distribution):

- `Q2 > 1 GeV2`
- `W > 2 GeV`
- `0.3 < zh < 0.7`



Work done by Gregory show during a collaboration meeting (March 2024).

Forward detector reconstruct “false photons” (MAGENTA and TEAL).



- RG-D analysis:
 - Understand the lc vs TA dependence
 - Finish the event mixing method (difference at low invariant mass)
 - Reproduce the plot show in the previous slide.
 - Extract the multiplicity ratio, pT broadening and asymmetry
- ALERT:
 - Nathan give me some comments about my pull request to improve it.
- Whitney ask me if I can work on the ALERT software to create a new engine ALERT were the track finding and track fitting is done. And where we can use the information from both AHDC and ATOF.
- Shifts